GG-B-821

MAY 14, 1948

FEDERAL STANDARD STOCK CATALOG

Section IV

(Part 5)

FEDERAL SPECIFICATION

FOR

BURS; DENTAL

This specification was approved on the above date by the Director, Bureau of Federal Supply, for the use of all departments and establishments of the Government, and shall become effective not later than August 15, 1948, for use in procurement initiated on or after that date. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE SPECIFICATIONS

A–1. The following Federal Specification, of the issue in effect on date of invitation for bids, shall form a part of this specification:

QQ–M–151—Metals; General Specification for Inspection of.

A–2. For specifications and publications applicable only to individual departments, see section H.

B. TYPES, CLASSES, AND SIZES

B–1. *Types.*—Dental burs covered by this specification shall be of the following types, as specified in the invitation for bids:

Type I.—Excavating burs.
Type II.—Finishing burs.
Type III.—Surgical burs.

B–2. *Classes.*—Dental burs covered by this specification shall be of the following classes, as specified in the invitation for bids:

Class A.—Burs for angle handpiece.
Class S.—Burs for straight handpiece.

B–3. *Sizes.*—Dental burs covered by this specification shall be of the sizes shown in tables I, II, and III, as specified in the invitation for bids.

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—The burs shall be made of an alloy steel containing not less than 1 percent carbon and 1 percent tungsten. The fractured surface of hardened steel shall show a fine grain, silky structure.

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C–2. *Workmanship.*—The burs shall be symmetrically made, smooth, and properly finished. The blades shall be properly cut and sharp.

D. GENERAL REQUIREMENTS

D–1. *Hardness.*—

D–1a. *Scratch test.*—When tested as described in paragraph F–3b(1), the burs shall scratch the glass and the blades shall not turn back.

D–1b. *Vickers test.*—When tested as described in paragraph F–3b(2), the hardness of the bur heads shall be not less than Number 750 on the Vickers machine.

D–2. *Eccentricity.*—When tested as described in paragraph F–3c, the eccentricity shall be not more than 0.003 inch over-all.

D–3. *Mandrels.*—The diameter of the mandrels at the shank end shall be 0.0924 ± 0.0005 inch.

D–4. *Marking.*—Each bur shank shall be stamped with the name or registered trade-mark of the manufacturer.

E. DETAIL REQUIREMENTS

E–1. *Type I.*—Type I dental burs shall be of the excavating type.

E–2. *Type II.*—Type II dental burs shall be of the finishing type.

E–3. *Type III.*—Type III dental burs shall be of the surgical type. They shall have three blades the angle of which shall be 8° to 10° right-hand thread.

E–4. *Class A.*—Class A burs shall be for use with angle handpieces. The shank end of the mandrel shall be cut away and notched as shown in figure 1, so that it will fit the standard dental angle handpiece.

E–5. *Class S.*—Class S burs shall be for use with straight handpieces. The shank end of the mandrel shall fit the standard dental straight handpiece as shown in figure 1.

E–6. *Shapes and dimensions.*—Dental burs shall be of the shapes and dimensions shown in tables I, II, and III for the various types, classes, and sizes, and shall have the number of blades shown therein.

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TABLE I.—*Type I, excavating burs.*

Size No.	Class	Shape	Number of blades	Dimensions			
				Over-all length ±0.010 inch		Head dimensions	
				Class A	Class S	Diameter[1] ±0.002 inch	Length, minimum
				Inch	*Inch*	*Inch*	*Inch*
000	A, S	Round	6	0.875	1.750	0.020	
½	A, S	do	6	0.875	1.750	0.025	
½SN	A	do	6	0.725		0.025	
1	A, S	do	6	0.875	1.750	0.032	
2	A, S	do	6	0.875	1.750	0.039	
2SN	A	do	6	0.725		0.039	
3	A, S	do	6	0.875	1.750	0.047	
4	A, S	do	8	0.875	1.750	0.055	
4SN	A	do	8	0.725		0.055	
5	A, S	do	8	0.875	1.750	0.063	
6	A, S	do	8	0.875	1.750	0.072	
6SN	A	do	8	0.725		0.072	
7	A, S	do	8	0.875	1.750	0.081	

[1] Designates diameter at largest place. SN = Short neck.

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TABLE I.—*Type I, excavating burs (cont'd).*

Size No.	Class	Shape	Number of blades	Dimensions			
				Over-all length ±0.010 inch		Head dimensions	
				Class A	Class S	Diameter[1] ±0.002 inch	Length, minimum
				Inch	*Inch*	*Inch*	*Inch*
8	A, S	Round	8–10	0.875	1.750	0.090	.....
9	A, S	do	10	0.875	1.750	0.099	.....
11	A, S	do	10	0.875	1.750	0.119	.....
12	A, S	Wheel	8	0.875	1.750	0.033	0.012
14	A, S	do	8	0.875	1.750	0.048	0.016
33½	A, S	Inverted cone	8	0.875	1.750	0.025	0.016
33½SN	A	do	8	0.725	.....	0.025	0.016
34	A, S	do	8	0.875	1.750	0.032	0.028
35	A, S	do	8	0.875	1.750	0.039	0.033
35SN	A	do	8	0.725	.....	0.039	0.033
36	A, S	do	8	0.875	1.750	0.047	0.040
37	A, S	do	8	0.875	1.750	0.055	0.046
37SN	A	do	8	0.725	.....	0.055	0.046

[1] Designates diameter at largest place. SN = Short neck.

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TABLE I.—*Type I, excavating burs (cont'd).*

Size No.	Class	Shape	Number of blades	Dimensions			
				Over-all length ±0.010 inch		Head dimensions	
				Class A	Class S	Diameter[1] ±0.002 inch	Length, minimum
				Inch	*Inch*	*Inch*	*Inch*
38	A, S	Inverted cone	8	0.875	1.750	0.063	0.053
39	A, S	do	8	0.875	1.750	0.072	0.060
39SN	A	do	8	0.725	.....	0.072	0.060
40	A, S	do	8	0.875	1.750	0.081	0.068
56	A	Plain fissure	6-8	0.875	.....	0.032	0.118
	S	do	6-8	.....	1.750	0.032	0.138
57	A	do	6-8	0.875	.....	0.039	0.138
	S	do	6-8	.....	1.750	0.039	0.158
59	A	do	6-8	0.875	.....	0.055	0.176
	S	do	6-8	.....	1.750	0.055	0.193
502	A, S	Round, crosscut	6	0.875	1.750	0.039	.....
504	A, S	do	8	0.875	1.750	0.055	.....
556	A	Straight fissure	6-8	0.875	.....	0.032	0.118
	S	do	6-8	.....	1.750	0.032	0.138
557	A	Straight fissure, crosscut	6-8	0.875	.....	0.039	0.138

[1] Designates diameter at largest place. SN = Short neck.

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TABLE I.—*Type I, excavating burs (cont'd).*

Size No.	Class	Shape	Number of blades	Dimensions			
				Over-all length ±0.010 inch		Head dimensions	
				Class A	Class S	Diameter[1] ±0.002 inch	Length, minimum
				Inch	*Inch*	*Inch*	*Inch*
557	S	Stroight fissure, crosscut	6–8	.....	1.750	0.039	0.158
557SN	A	do	6–8	0.725	.....	0.039	0.111
558	A	do	6–8	0.875	.....	0.047	0.138
	S	do	6–8	.....	1.750	0.047	0.178
558SN	A	do	6–8	0.725	.....	0.047	0.111
559	A	Straight fissure, crosscut, flat end	6–8	0.875	.....	0.055	0.178
	S	do	6–8	.....	1.750	0.055	0.193
559SN	A	do	6–8	0.725	.....	0.055	0.111
560	A	Straight fissure, crosscut	6–8	0.875	.....	0.063	0.178
	S	do	6–8	.....	1.750	0.063	0.208
562	A	Straight fissure	6–8	0.875	.....	0.081	0.178
	S	do	6–8	.....	1.750	0.081	0.235
563	A	do	8–10	0.875	.....	0.090	0.208
	S	do	8–10	.....	1.750	0.090	0.258
600	A	Tapered fissure, fine cut, inlay	8–14	0.875	.....	[2]0.043	0.138
	S	do	8–14	.....	1.750	[2]0.043	0.158

[1] Designates diameter at largest place. [2] Permissible tolerance of plus or minus 0.005 inch. SN = Short neck.

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TABLE I.—*Type I, excavating burs (cont'd).*

Size No.	Class	Shape	Number of blades	Dimensions			
				Over-all length ±0.010 inch		Head dimensions	
				Class A	Class S	Diameter[1] ±0.002 inch	Length, minimum
				Inch	*Inch*	*Inch*	*Inch*
601	A	Tapered fissure, fine cut, inlay	8–14	0.875	.....	[3]0.055	0.138
	S	do	8–14	.....	1.750	[3]0.055	0.178
603	A	Tapered fissure, fine cut, inlay, crosscut	8–14	0.875	.....	[2]0.043	0.138
	S	do	8–14	.....	1.750	[2]0.043	0.158
604	A	do	8–14	0.875	.....	[3]0.055	0.138
	S	do	8–14	.....	1.750	[3]0.055	0.178
700	A	Tapered fissure, coarse cut, crosscut	6–8	0.875	.....	0.039	0.137
	S	do	6–8	.....	1.750	0.039	0.158
700L	A, S	Tapered fissure, coarse cut	8	0.9375	1.170	0.039	0.240
701	A	Tapered fissure, coarse cut, crosscut	6–8	0.875	.....	0.047	0.138
	S	do	6–8	.....	1.750	0.047	0.178
701SN	A	do	6–8	0.725	.....	0.047	0.111
701L	A, S	Tapered fissure, coarse cut	8	0.9375	1.170	0.047	0.240
702	A	Tapered fissure, coarse cut, crosscut	6–8	0.875	.....	0.063	0.178
	S	do	6–8	.....	1.750	0.063	0.208
702SN	A	do	6–8	0.725	.....	0.063	0.111

[1] Designates diameter at largest place.
[2] Permissible tolerance of plus or minus 0.005 inch.
[3] Permissible tolerance of plus or minus 0.009 inch.

SN = Short neck.
L = Long neck.

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TABLE I.—*Type I, excavating burs (cont'd).*

Size No.	Class	Shape	Number of blades	Dimensions			
				Over-all length ±0.010 inch		Head dimensions	
				Class A	Class S	Diameter[1] ±0.002 inch	Length, minimum
				Inch	*Inch*	*Inch*	*Inch*
703	A	Tapered fissure, coarse cut	6–8	0.875	.....	0.081	0.178
	S	do	6–8	.....	1.750	0.081	0.238
957	A	End cutting	6–8	0.875	.....	0.039	0.138
	S	do	6–8	.....	1.750	0.039	0.168
958	A	do	6–8	0.875	.....	0.046	0.138
	S	do	6–8	.....	1.750	0.046	0.178
959	A	do	6–8	0.875	.....	0.055	0.176
	S	do	6–8	.....	1.750	0.055	0.193

[1] Designates diameter at largest place.

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TABLE II.—*Type II, finishing burs.*

Size No.	Class	Shape	Number of blades	Dimensions			
				Over-all length ±0.010 inch		Head dimensions	
				Class A	Class S	Diameter[1] ±0.002 inch	Length, minimum
				Inch	*Inch*	*Inch*	*Inch*
C	A, S	Round	16	0.875	1.750	0.055	.....
D	A, S	do	16	0.875	1.750	0.063	.....
200	A, S	do	20	0.875	1.750	0.081	.....
202	A, S	do	24	0.875	1.750	0.119	.....
218	A, S	Oval	20	0.875	1.750	0.081	0.038
219	A, S	do	20	0.875	1.750	0.097	0.055
224	A, S	Bud	20	0.875	1.750	0.081	0.123
231	A, S	Pear	20	0.875	1.750	0.099	0.103
242	A, S	Sugar loaf	10	0.875	1.750	0.039	0.170
245	A, S	do	16	0.875	1.750	0.063	0.211

[1] Designates diameter at largest place.

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TABLE III.—*Type III, surgical burs.*

Size No.	Class	Shape	Number of blades	Dimensions				
				Over-all length ±0.010 inch		Head dimensions		
				Class A	Class S	Diameter[1] ±0.004 inch	Length ±0.002 inch
				Inch	*Inch*	*Inch*	*Inch*
41	A, S	"Henahan"	3	1.125	2.50	0.072	0.150

[1] Designates diameter at largest place.

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F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Sampling.*—Unless otherwise specified, at least one package of six burs of each type, class, and size, shall be taken at random from each lot of 500 packages or fraction thereof delivered, for inspection and tests.

F–2. *Inspection.*—Samples shall be inspected for material, workmanship, markings, dimensions, and shapes.

F–3. *Tests.*—

F–3a. *Dimensions.*—Measurements shall be made with a suitable toolmaker's microscope or other instrument of equal accuracy.

F–3b. *Hardness.*—

F–3b(1). *Scratch test.*—The burs shall be held in the hand and the heads drawn over plate glass under hand pressure. After the test the bur head and the plate glass shall be examined to determine conformance to the requirements of paragraph D–1a.

F–3b(2). *Vickers test.*—If desired by the purchasing agency, tests for hardness shall be conducted on the Vickers machine as described in Federal Specification QQ–M–151. Samples shall be prepared for tests as described in Federal Specification QQ–M–151. The preparation of the specimen shall result in two parallel surfaces on the head of the bur, and extreme care to prevent any heating of the bur shall be taken in preparing the specimen by applying sufficient coolant.

F–3c. *Eccentricity.*—Eccentricity shall be determined by a shadowgraph or by rotating the bur in a vee block by means of a round file. When the latter method is used, the eccentricity at the neck of the bur shall be measured by means of a dial gage, and the total movement shown on the gage shall be the over-all eccentricity.

F–3d. *Material.*—Conformance to the requirements of paragraph C–1 shall be determined by chemical analysis or by a spectrograph.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging.*—Burs shall be coated with a rust-preventive compound and packaged six of one type, class, and size in a plastic holder, in a moistureproof package. Each package of six burs shall have a transparent window to render heads of burs visible.

G–2. *Packing.*—The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

G–3. *Marking.*—

G–3a. *Packages.*—Unless otherwise specified, packages shall be marked with the name, type, class, and size of material, the quantity contained therein, and the names of the contractor and manufacturer.

G–3b. *Shipping containers.*—Unless otherwise specified, each shipping container shall be marked with the name and type of material as defined by the contract or order under which the shipment is made, the quantity of each type contained therein, the

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name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS

H–1. The following specifications and other publications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

H–2. *Army and Navy.*—

H–2a. *Applicable specifications and other publications.*—

Joint Army-Navy Specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

JAN–B–107—Boxes, Wood, Wirebound (Overseas Type).

JAN–P–116—Packaging and Packing for Overseas Shipment—Preservation, Methods Of.

JAN–P–120—Packaging and Packing for Overseas Shipment — Cartons, Folding, Paperboard.

JAN–P–125—Packaging and Packing for Overseas Shipment — Barrier-Materials, Waterproof, Flexible.

JAN–P–133—Packaging and Packing for Overseas Shipment—Boxes, Set-Up, Paperboard.

JAN–P–140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner.

U. S. Army Specification:

100–2—Standard Specification for Marking Shipments by Contractors.[1]

Navy Department Specifications:

General Specifications for Inspection of Material.[2]

52C18—Compound, Rust-Preventive, Thin-Film (Polar Type).

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.[2]

H–2a(1). Copies of specifications required for Army purchases may be obtained upon application to the procuring agency. Both the title and identifying symbol or number should be stipulated when requesting copies.

H–2a(2). Copies of Federal and Joint Army-Navy specifications required for Navy purchases, Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained without cost upon application to the Bureau of Supplies and Accounts,

[1] Applicable to Army purchases only.
[2] Applicable to Navy purchases only.

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Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol or number should be stipulated when requesting copies of specifications.

H–2b. *Packaging.*—

H–2b(1). *Unit packages.*—Six burs shall constitute one unit. The burs shall be coated with rust-preventive compound conforming to the requirements for grade III of Navy Department Specification 52C18. Six burs so treated shall be inserted into a plastic holder. The burs shall be further protected with a paperboard cover which shall be overwrapped with moisture-resistant cellulose acetate and heat sealed.

H–2b(2). *Intermediate packages.*—

H–2b(2)a. *Domestic.*—Twelve unit packages, 72 burs, shall be further packaged in a folding paperboard carton or set-up paperboard box of appropriate size, conforming to the requirements of Joint Army-Navy Specifications JAN–P–120 and JAN–P–133, respectively.

H–2b(2)b. *Overseas.*—Twelve unit packages, 72 burs, shall be packaged as described in paragraph H–2b(2)a. In addition, each carton shall be packaged in accordance with method IA of Joint Army-Navy Specification JAN–P–116, except that no special cleaning or preservative shall be required; technique shall conform to method IA2 or IA3 and appendix IV.

H–2c. *Packing.*—

H–2c(1). *Domestic.*—Unless otherwise specified, the following quantities shall be packed for type 2 load in a cleated plywood box, nailed wood box, or wire-bound wood box, conforming to the requirements of Joint Army-Navy Specifications JAN–P–105, JAN–P–106, and JAN–B–107, respectively.

Type	Class	Number of intermediate packages to shipping container	Number of unit packages to shipping container
I	A	192	2304
I	S	96	1152
II	A and S	96	1152
III	A and S	96	1152

H–2c(2). *Overseas.*—Unless otherwise specified, the commodity shall be packed as described in paragraph H–2c(1). Shipping containers shall be lined with a sealed, waterproof bag, or its equivalent, made from material conforming to the requirements of Joint Army-Navy Specification JAN–P–125 for case liners; seams and closures shall be sealed with water-resistant adhesive conforming to the requirements of Joint Army-Navy Specification JAN–P–140.

H–2c(3). *Packing variation.*—If the number of units in any shipment is not the exact multiple of the number of units specified

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to be contained in each shipping container, the units not constituting the nearest such multiple shall be packed in an exterior container of suitable size and design and so constructed as to insure safe delivery to specified destination.

H–2d. *Marking.*—

H–2d(1). *Packages.*—Each intermediate package shall be marked with the stock number, nomenclature, unit, quantity, and name of manufacturer.

H–2d(2). *Shipping containers.*—

H–2d(2)a. *Army.*—Each shipping container shall be marked in accordance with the requirements of U. S. Army Specification 100–2. The legend "METHOD IA PKG" shall be marked on the shipping container when items are processed as indicated in paragraph H–2b(2)b.

H–2d(2)b. *Navy.*—Each shipping container shall be marked in accordance with the requirements of the Navy Shipment Marking Handbook.

H–2e. *Ordering information.*—Requests, requisitions, schedules, and contracts or orders should specify whether the subject commodity is to be packaged, packed, and shipping containers marked for domestic or overseas shipment. (See pars. H–2b(2), H–2c, and H–2d(2).)

NAVY: M S

I. NOTES

I–1. Purchasers should specify type, class, and size required, and should exercise any desired options offered herein. (See pars. B–1, B–2, B–3, F–1, G–3, and H–2e.)

I–2. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

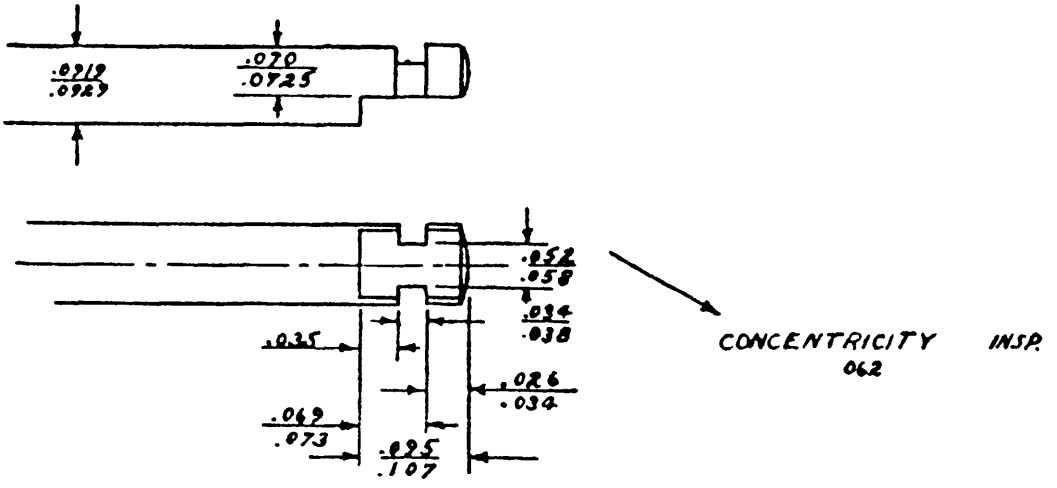
I–3. Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

I–4. An Index of Federal Specifications may be purchased as noted in the paragraph below, price to be obtained from the Superintendent of Documents.

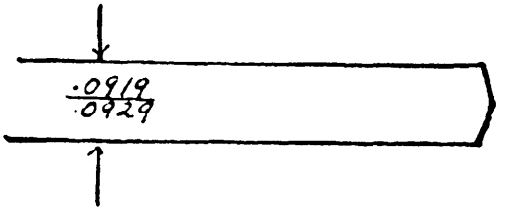
I–5. Copies of this specification and of Federal Specification QQ–M–151 may be obtained upon application, accompanied by postal note, money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price 5 cents each.

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Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.



CLASS A



CLASS S

FIGURE 1.—Shank end of mandrels.

For sale by the Superintendent of Documents, Washington 25, D. C.
Price 5 cents.

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